

Verifier-Gated Discovery Lab

Agents propose, Omnigent orchestrates, only a code verifier accepts. Commit 8c47757

Challenge and users

Agentic scientific discovery (Databricks challenge). Users: researchers who need results they can trust without re-checking every line.

Tools

Omnigent (orchestration, policies), Claude via the Claude Code CLI, Python verifiers (sympy, python-flint, scipy), MCP server, LaTeX.

What worked

The verifier-gated loop: 3 certified and 1 rejected claims in one run; a surprise reopened an assumption and changed the plan; the family classification moved to 50 / 11 / 27.

What was hard

Making the agents' claims exactly as strong as the evidence; keeping tolerances inside the verifier; parallel agents writing one state file (fixed with a file lock).

Time use

Recorded run 19 min, of which 9 min waiting for human approval; agents/LLM at least 195 s, verifier at most 57 s, experiments at most 196 s.

Results

11 exactly certified violations of the proofreading bound among 88 topologies, 8 of them new in the recorded run; trust benchmark: lab 0.0% false vs. Claude alone 36.1%.

Measured speedup

Replay: 4.6x vs. random search (95% CI 2.6 to 7.5, $p = 0.003$); vs. no feedback 1.2x (not significant). Result-to-decision latency median 10 s ($n = 4$). A stopwatch baseline by the team is pending; no comparison with a real laboratory.

Next 24 h

Decide the 27 open topologies (cycle decomposition for proofs, wider counterexample search); stopwatch baselines; second domain.